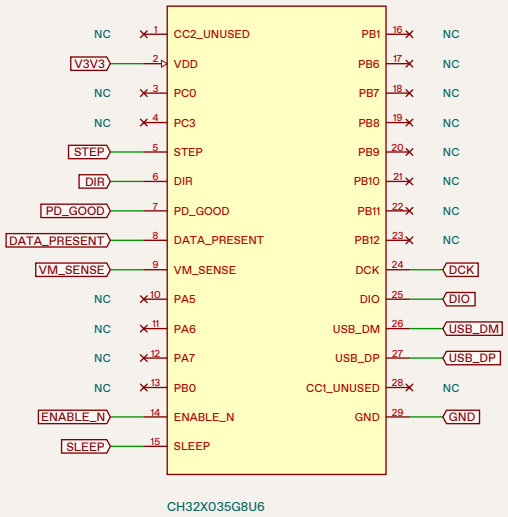
# NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 1/9 - U_MCU

## U_MCU




CH32X035G8U6

## COMPONENT PURPOSE

### U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal clock avoids an external crystal.

### U_PD - CH224K

USB-C PD sink requests a 15 V motor supply.
PG reports a completed power negotiation.

### U_LDO - HT7533-1

Creates 3.3 V for MCU and driver logic.
Diode OR accepts either USB data power or PD.

### U_DRV - A4988

Two current-regulated bridges drive the coils.
STEP/DIR interface; fixed 1/16 microstepping.
0.24 ohm sense resistors set phase current.

### U_ESD - USBLC6-2SC6

Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the USB protection diodes.

Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

# NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 2/9 - U_DRV

## COMPONENT PURPOSE

### U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal clock avoids an external crystal.

### U_PD - CH224K

USB-C PD sink requests a 15 V motor supply.
PG reports a completed power negotiation.

### U_LDO - HT7533-1

Creates 3.3 V for MCU and driver logic.
Diode OR accepts either USB data power or PD.

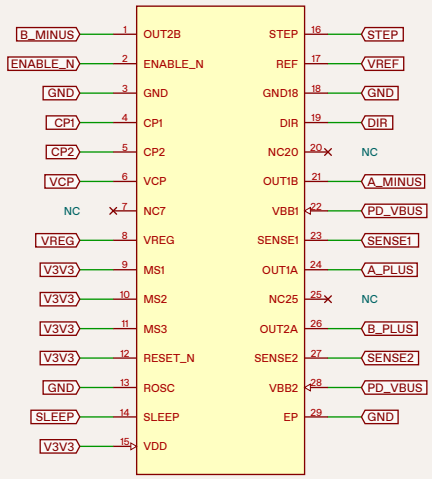
### U_DRV - A4988

Two current-regulated bridges drive the coils.
STEP/DIR interface; fixed 1/16 microstepping.
0.24 ohm sense resistors set phase current.

### U_ESD - USBLC6-2SC6

Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the USB protection diodes.

U_DRV




A4988SETTR-T

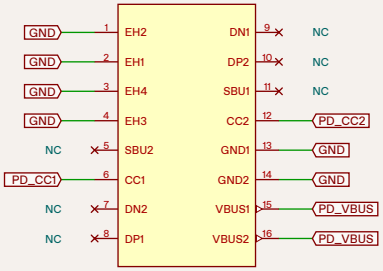
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Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

# NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

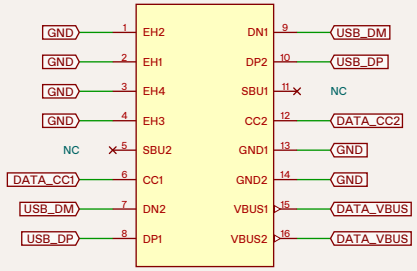
SHEET 3/9 - J_PD, J_DATA, U_PD, R_PD, C_PD, R_PG

## J_PD



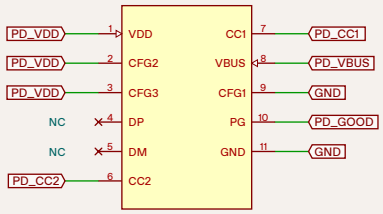

TYPE-C-31-M-12

## J_DATA




TYPE-C-31-M-12

## U_PD

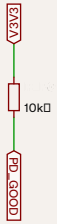



CH224K










Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

## COMPONENT PURPOSE

### U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal clock avoids an external crystal.

### U_PD - CH224K

USB-C PD sink requests a 15 V motor supply.
PG reports a completed power negotiation.

### U_LDO - HT7533-1

Creates 3.3 V for MCU and driver logic.
Diode OR accepts either USB data power or PD.

### U_DRV - A4988

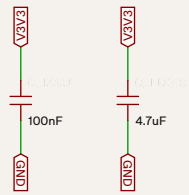
Two current-regulated bridges drive the coils.
STEP/DIR interface; fixed 1/16 microstepping.
0.24 ohm sense resistors set phase current.

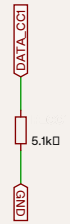
### U_ESD - USBLC6-2SC6

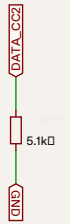
Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the USB protection diodes.

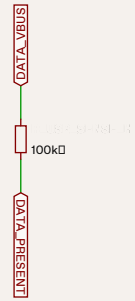
# NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

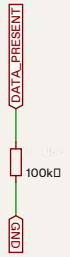
SHEET 4/9 - C_MCU, C_LOGIC, R_CC1, R_CC2, R_USB_SENSE_H, R_USB_SENSE_L




## COMPONENT PURPOSE

### U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal clock avoids an external crystal.

### U_PD - CH224K

USB-C PD sink requests a 15 V motor supply.
PG reports a completed power negotiation.

### U_LDO - HT7533-1

Creates 3.3 V for MCU and driver logic.
Diode OR accepts either USB data power or PD.

### U_DRV - A4988

Two current-regulated bridges drive the coils.
STEP/DIR interface; fixed 1/16 microstepping.
0.24 ohm sense resistors set phase current.

### U_ESD - USBLC6-2SC6

Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the USB protection diodes.

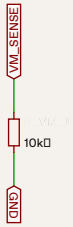
Named nets connect across all pages. NC = intentionally unconnected pin.

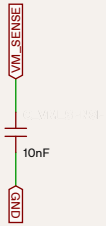
PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

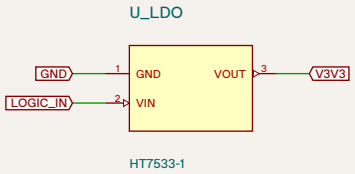
# NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 5/9 - R_VM_H, R_VM_L, C_VM_SENSE, U_LDO, D_PD, D_DATA










## COMPONENT PURPOSE

### U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal clock avoids an external crystal.

### U_PD - CH224K

USB-C PD sink requests a 15 V motor supply.
PG reports a completed power negotiation.

### U_LDO - HT7533-1

Creates 3.3 V for MCU and driver logic.
Diode OR accepts either USB data power or PD.

### U_DRV - A4988

Two current-regulated bridges drive the coils.
STEP/DIR interface; fixed 1/16 microstepping.
0.24 ohm sense resistors set phase current.

### U_ESD - USBLC6-2SC6

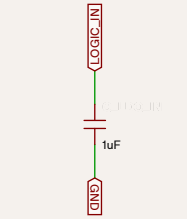
Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the USB protection diodes.

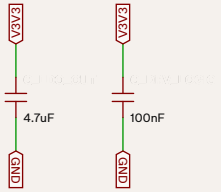
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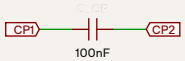
PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

# NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 6/9 - C_LDO_IN, C_LDO_OUT, C_CP, C_VCP, C_VREG, C_DRV_LOGIC










## COMPONENT PURPOSE

### U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal clock avoids an external crystal.

### U_PD - CH224K

USB-C PD sink requests a 15 V motor supply.
PG reports a completed power negotiation.

### U_LDO - HT7533-1

Creates 3.3 V for MCU and driver logic.
Diode OR accepts either USB data power or PD.

### U_DRV - A4988

Two current-regulated bridges drive the coils.
STEP/DIR interface; fixed 1/16 microstepping.
0.24 ohm sense resistors set phase current.

### U_ESD - USBLC6-2SC6

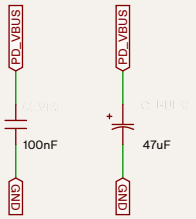
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VBUS pin references the USB protection diodes.

Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

# NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 7/9 - C_VM, C_BULK, R_SA, R_SB, R_REF_H, R_REF_L
















## COMPONENT PURPOSE

### U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal clock avoids an external crystal.

### U_PD - CH224K

USB-C PD sink requests a 15 V motor supply.
PG reports a completed power negotiation.

### U_LDO - HT7533-1

Creates 3.3 V for MCU and driver logic.
Diode OR accepts either USB data power or PD.

### U_DRV - A4988

Two current-regulated bridges drive the coils.
STEP/DIR interface; fixed 1/16 microstepping.
0.24 ohm sense resistors set phase current.

### U_ESD - USBLC6-2SC6

Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the USB protection diodes.

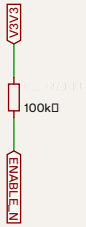
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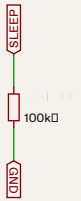
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# NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

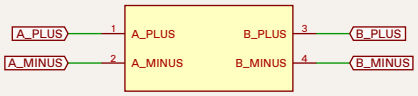
SHEET 8/9 - C_REF, R_ENABLE, R_SLEEP, J_MOTOR, J_DEBUG, U_ESD




J_MOTOR



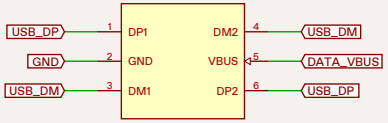

B4B-PH-K-S(LF)(SN)

J_DEBUG




PCB solder interface

U_ESD




USBLC6-2SC6

## COMPONENT PURPOSE

### U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal clock avoids an external crystal.

### U_PD - CH224K

USB-C PD sink requests a 15 V motor supply.
PG reports a completed power negotiation.

### U_LDO - HT7533-1

Creates 3.3 V for MCU and driver logic.
Diode OR accepts either USB data power or PD.

### U_DRV - A4988

Two current-regulated bridges drive the coils.
STEP/DIR interface; fixed 1/16 microstepping.
0.24 ohm sense resistors set phase current.

### U_ESD - USBLC6-2SC6

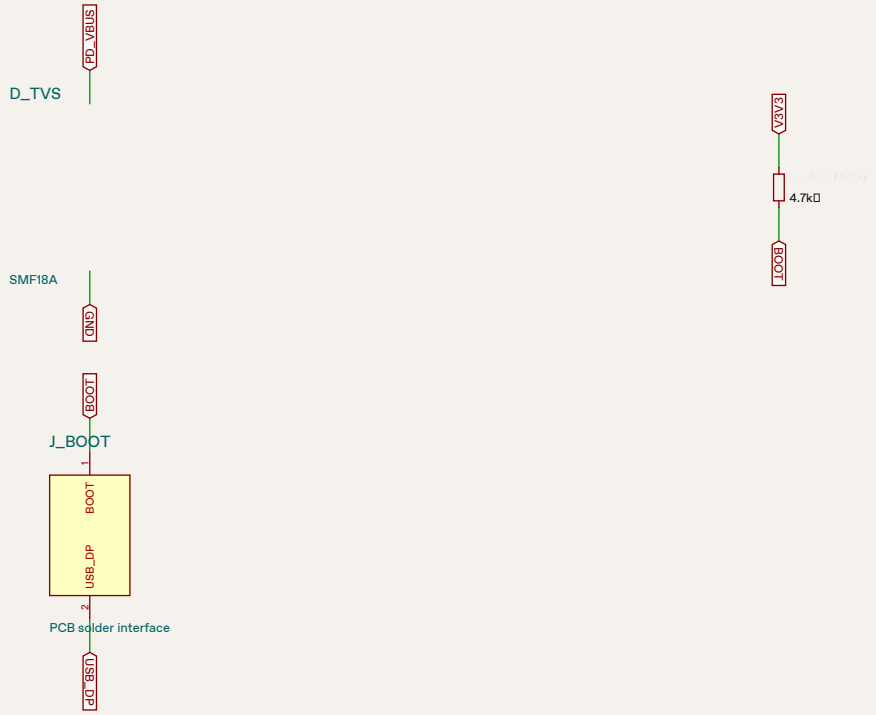
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Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

# NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 9/9 - D_TVS, R_USB_BOOT, J_BOOT




## COMPONENT PURPOSE

### U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal clock avoids an external crystal.

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Creates 3.3 V for MCU and driver logic.
Diode OR accepts either USB data power or PD.

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